Pre-Calculus 12 - Polynomial Functions



— SINE TIMORE AUT FAVORE —

ST. GEORGE'S SCHOOL

Name : _____

Teacher's name: _____

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1 Characteristics of Polynomials

Polynomial Function

A **polynomial function** has the form

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_2 x^2 + a_1 x + a_0$$

where,

- n is a whole number
- x is a variable
- the coefficients a_n to a_0 are real numbers

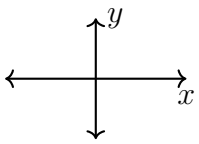
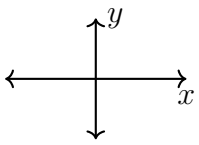
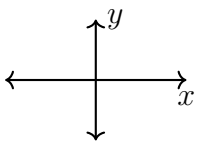
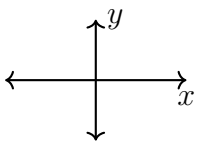
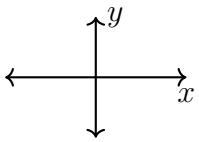
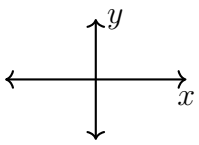
Example 1.1. Explain why the following are **NOT** polynomial functions.

Function	Reason
$y = \frac{1}{x} - 2x^2 - x - 1$	
$y = \sqrt{4x} + 8x + 2$	
$y = x^3 + x^2 - x^{-1}$	

Example 1.2. Is $f(x) = 4$ a polynomial function? Explain why or why not.

Key Terms

Term	Definition	Example $y = -5x^4 + 3x^2 + 6$
Degree	The greatest exponent of the variable	
Leading Coefficient	The numerical coefficient of the highest degree variable	
Constant	The numerical value of the term without a variable (ie. x^0)	
End behaviour	The behaviour of the y-values as $\|x\|$ becomes very large (as x approaches $\pm\infty$)	

Function	Sketch of Graph	Degree	Leading Coefficient	End Behaviour	Max # of Zeros	y-int
$y = x - 3$						
$y = -x^2 - 3x + 2$						
$y = 2x^3 - 3x^2 - x + 1$						
$y = -x^3 - 3x^2 + 4x + 15$						
$y = x^4 - 2x^3 - x^2 + 2x$						
$y = -2x^5 + 5x^3 - x$						

Characteristics of Polynomials

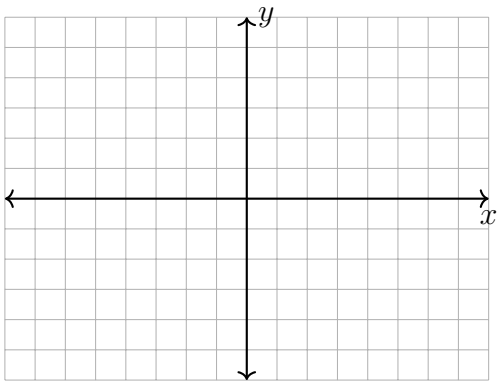
- The degree of a polynomial indicates the **maximum** number of x-intercepts (also called zeros or roots).
- If the degree is **odd**, the ends will point in **opposite** directions.
- If the degree is **even**, the ends will point in the **same** direction.
- The sign of the leading coefficient indicates if the function has been reflected over the x-axis.
- The constant indicates the y-intercept.

Example 1.3. Fill in the blank for the following statements.

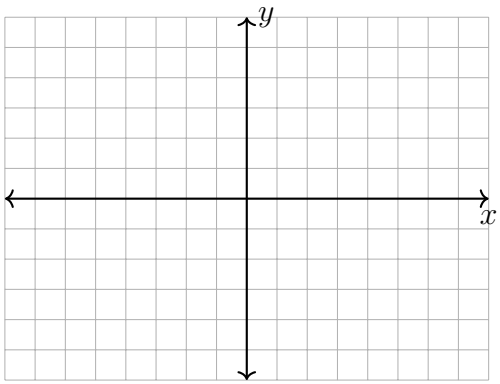
- $y = x^5 + x^2 - 4$ has a maximum of ___ x-intercepts.
- $y = x - 4$ has a maximum of ___ x-intercepts.
- $y = -4$ has ___ x-intercepts.

Example 1.4. For each of the following, sketch a polynomial function with the given characteristics.

a) A polynomial function with degree 3, a positive leading coefficient and 2 x-intercepts.



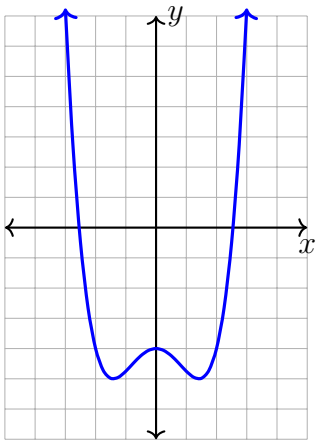
b) A polynomial function with degree 4, a negative leading coefficient and 4 x-intercepts.



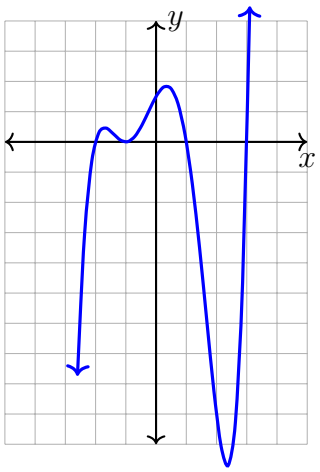
Example 1.5. For each graph of a polynomial function,

- determine whether the function has an odd degree or even degree
- determine whether the leading coefficient is positive or negative
- state the number of x-intercepts
- state the domain and range

a)



b)



Example 1.6. For each function, use the degree and the sign of the leading coefficient to describe the end behaviour of its graph. State the possible number of x-intercepts and the value of the y-intercept.

a) $g(x) = 4x^5 - x^3 + 3x^2 - 6x + 2$

b) $y = -x^4 - 2x^5 + x^3 - 3x^2 + x$

c) $h(x) = x - 7x^3 - 6$

d) $y = 3x^5 - 2x^4 + 5x^3 - x^2 + x + 3$

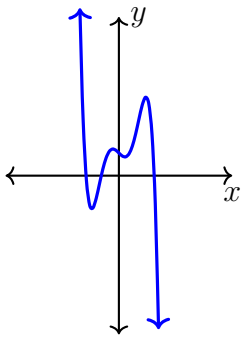
Practice Exercises

1. Match each function to its corresponding graph without the use of a calculator.

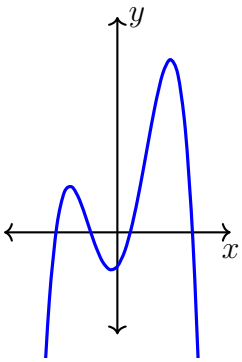
   a) $f(x) = 2x^3 - 4x^2 + x + 2$ b) $g(x) = -x^4 + 10x^2 + 5x - 6$

   c) $h(x) = -2x^5 + 5x^3 - x + 1$ d) $p(x) = x^4 - 5x^3 + 16$

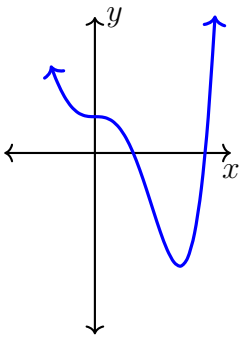
A



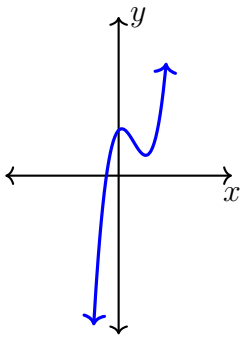
B



C



D



2. Complete the table for each polynomial function

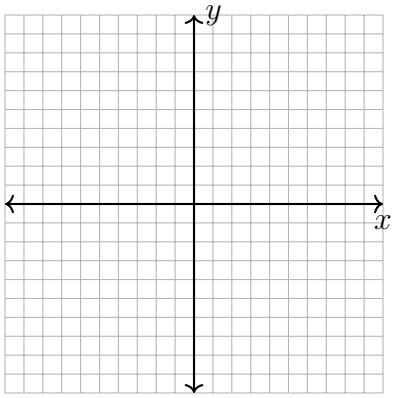
Polynomial Function	Degree	Type	Leading Coefficient	Constant Term
$f(x) = 6x^3 - 5x^2 + 2x - 8$	3			
$y = -2x^5 + 5x^3 + x^2 + 1$		Quintic		
$g(x) = x^3 - 7x^4$				
$p(x) = 10x - 9$				
$y = -0.5x^2 + 4x + 3$				
$h(x) = 3x^4 - 8x^3 + x^2 + 2$				
$y = -5$		Constant		

3. For each function, use the degree and the sign of the leading coefficient to describe the end behaviour of its graph. State the possible number of x-intercepts and the value of the y-intercept.

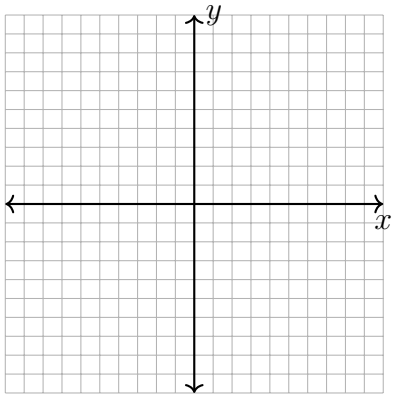
   a) $y = -x^4 - 2x^5 + x^3 - 3x^2 + x$ b) $p(x) = 5x^4 - 6x - 1$

4. On each set of axes, sketch a polynomial function with the given characteristics.

   a) A polynomial function with degree 5, a positive leading coefficient, and 2 x-intercepts.



   b) A polynomial function with degree 4, a negative leading coefficient, and 4 x-intercepts.



5. Identify which of the following is a polynomial function

   a) $h(x) = 2 - \sqrt{x}$ b) $y = 3x + 1$ c) $f(x) = 2x^2 + 3^x$

   d) $g(x) = 3x^4 - 7$ e) $p(x) = x^{-3} + x^2 + 3x$ f) $y = -4x^3 + 2x + 5$

   g) $y = -6$ h) $t(x) = 2x^4 - \dfrac{1}{x^{-2}}$

6. A demographer uses a polynomial function to predict the population, P, of a town t years from now. The function is $P(t) = t^4 - 20t^3 - 20t^2 + 1500t + 15000$. Assume the model can be used for the next 20 years.

   a) What is the current population of this town?

   b) What will the population of the town be in 10 years?

   c) When will the population of the town be approximately 24,000?

7. For each function, use the degree and the sign of the leading coefficient to describe the end behaviour of its graph. State the possible number of x-intercepts and the value of the y-intercept.

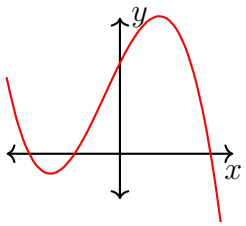
   a) $y = x(x + 1)^2(x + 4)^2$

   b) $y = -2x^2(x - 3)(x + 4)$

8. The graph of the polynomial is shown on the right, (where a, b, c, and d are constants). What must be true about a and d?

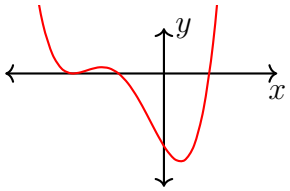
   $P(x) = ax^3 + bx^2 + cx + d$

   a) $a > 0, d \neq 0$
   b) $a > 0, d = 0$
   c) $a < 0, d \neq 0$
   d) $a < 0, d = 0$



9. The following graph represents the polynomial function $y = ax^4 + bx^3 + cx^2 + dx + e$. What conditions must be satisfied by a and e?

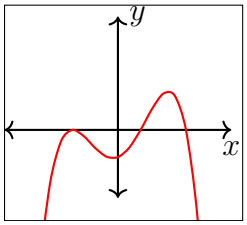
   a) $a < 0, e < 0$
   b) $a < 0, e > 0$
   c) $a > 0, e < 0$
   d) $a > 0, e > 0$



10. What is the minimum number of roots that a polynomial can have if its degree is 5?

    a) 0 b) 1 c) 3 d) 5

11. Karen makes the following statements about the graph of the polynomial function to the right



    i. The range of the function is $\{y \mid y \geq maximum\, y - value\}$
    ii. The constant term of the corresponding function is negative
    iii. There are 3 unique zeros
    iv. Leading coefficient of the polynomial is negative

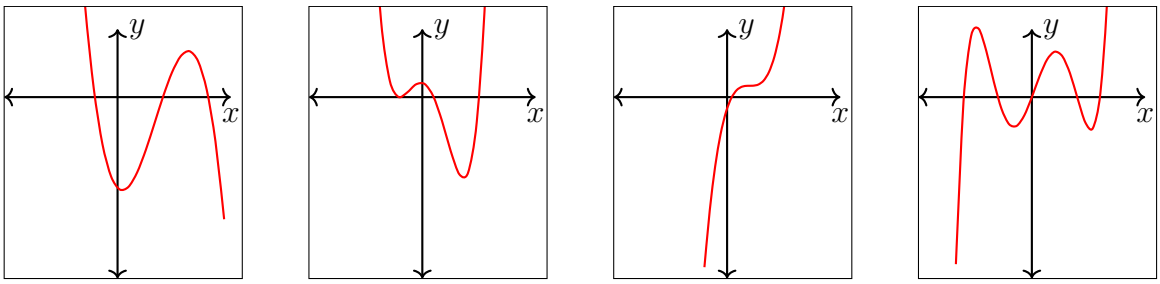
    Which of the statements are correct?

    a) i and ii b) ii and iii c) ii, iii, iv d) i, ii, iii, iv

12. Which of the following is a polynomial function?

    a) $f(x) = x^3 - x^2 + x - x^{-1}$ b) $f(x) = \sqrt{3x} + 4x$
    c) $f(x) = \dfrac{1}{x} + x^2$ when $x > -2$ d) $f(x) = -4$

13. Sam is presented with the following four partial graphs of polynomial functions, and is asked to select the graphs with an odd degree.



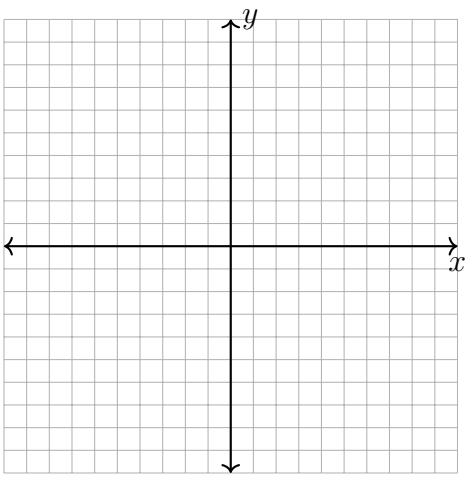
The number of graphs that Sam selects as having an odd degree is

a) 1 b) 2 c) 3 d) 4

2 The Remainder Theorem

Opener: Sketch $y = -2x^3 + 4x^2 - 1$

Identify the zeros, end behaviour, and domain and range:



Long Division

1. Write in long division form, including all degrees of the variable in the dividend in descending order - including $0x^n$ if necessary.
2. Divide the first term of the divisor into the first term of the dividend and record the answer as part of the quotient; line up the degree of the variables.
3. Multiply the quotient from Step 2 by each term in the divisor. Line the degrees up under the dividend.
4. Subtract like terms. The first term should subtract to zero.
5. Drop down the next term of the dividend and repeat Steps 2-4 as necessary.

Label the dividend, divisor, quotient and remainder for

$$\begin{array}{r} x+4 \\ x-2\overline{)\;x^2+2x+4} \\ \underline{-x^2+2x} \\ 4x+4 \\ \underline{-4x+8} \\ 12 \end{array}$$

Example 2.1. Divide the following using **long division**.

a) $x-2\overline{)\,3x^3-2x^2+1}$

b) $x+2\overline{)\,2x^3-4x^2+3x-6}$

Synthetic Division

Synthetic division is a shortcut for dividing polynomials and it works well when the divisor is of degree 1. Long division is still the best for all general polynomial division.
Below is **Example 2.1b** using synthetic division.

$$\begin{array}{r|cccc} & 2 & -4 & 3 & -6 \\ -2 & & -4 & 16 & -38 \\ \hline & 2 & -8 & 19 & -44 \end{array}$$

Example 2.2. Divide the following using **synthetic division**.

a) $(2x^3+4x^2-3x+4)\div(x-2)$

b) $(x^3+4x^2-x-6)\div(x+1)$

Remainder Theorem

If $P(x)$ is a polynomial, the remainder of $P(x) \div (x - a)$ is $P(a)$.

Example 2.3. Calculate the remainder when:

a) $4x^3 - 11x^2 + 12$ is divided by $x - 2$ b) $2x^3 - 3x^2 + x - 1$ is divided by $x + 1$

Example 2.4. When the polynomial $P(x) = x^3 - kx^2 + 17x + 6$ is divided by $x - 3$, the remainder is 12. What is the value of k?

Example 2.5. When the polynomial $P(x) = 3x^3 + mx^2 + nx - 7$ is divided by $x - 2$, the remainder is -3. When the polynomial is divided by $x + 1$ the remainder is -18. What are the values of m and n?

Practice Exercises

1. Use long division to evaluate the quotient and remainder for the following

   a) $23\overline{)\,546}$

   b) $1253 \div 43$

2. Use long division to divide $x^3 + 3x^2 - 2x + 5$ by $x + 1$. Express your result in the form $P(x) = d \cdot Q + r$

3. Divide the following using synthetic division. Express your solution using the division statement as shown above.

   a) $(2x^3 + x^2 - 27x - 36) \div (x + 3)$

   b) $(3x^4 + 2x^3 - 6x + 1) \div x$

   c) $(3x^4 - x^3 - 5) \div (x - 3)$

   d) $(-4x^4 + 11x - 7) \div (x - 3)$

4. Determine the remainder when each polynomial is divided by $(x-2)$

   a) $P(x) = 2x^3 + 3x^2 - 17x - 30$ b) $P(x) = x^3 + x^2 - 4x + 4$

5. Determine each remainder

   a) $(x^3 - x^2 - 2x - 1) \div (x + 2)$ b) $(2x^3 - 5x^2 - 13x + 2) \div (x - 4)$

   c) $(x^4 - 3x^2 - 5x + 2) \div (x - 2)$

6. For what value of m will the polynomial $P(x) = x^3 + 6x^2 + mx - 4$ have the same remainder when it is divided by $(x-1)$ **and** $(x+2)$?

7. If a polynomial $P(x)$ is divided by $(x+4)$, what is the remainder?

   a) $P(-4)$ b) $P(4)$ c) $P(x-4)$ d) $P(x+4)$

8. When $P(x)$ is divided by $(x-a)$, the remainder is b. Which of the following must be true?

   a) $P(-b) = a$ b) $P(b) = a$ c) $P(a) = b$ d) $P(-a) = b$

9. When a polynomial $P(x)$ is divided by $(x+3)$, the remainder is 4. Which point must be on the graph?

   a) $(-4, -3)$ b) $(-3, 4)$ c) $(4, -3)$ d) $(-3, -4)$

10. When $x^3 - x^2 + kx + 5$ is divided by $x + 2$, the remainder is 1. Find the value of k.

    a) -4 b) -1 c) 0 d) 4

11. Find a and b if $x + 1$ and $x + 2$ are both factors of $x^3 + ax + b$

    a) $a = -7, b = -6$
    b) $a = -7, b = 6$
    c) $a = 7, b = -6$
    d) $a = 7, b = 6$

12. If the polynomial $8x^3 + ax^2 + 16x + 3$ is divided by $4x - 3$, the quotient is $2x^2 + bx + 1$ and the remainder is 6. Find the values of a and b.

    a) $a = -22, b = -4$
    b) $a = -22, b = 4$
    c) $a = 22, b = -4$
    d) $a = 22, b = 4$

13. When x^3+ax+3 is divided by $x+2$, the remainder is equal to the square of the remainder when the same trinomial is divided by $x - 1$. Determine the value of a.

    a) $3, 7$
    b) $3, -7$
    c) $-4, 7$
    d) $-3, -7$

14. When the polynomial $ax^3 + bx^2 + cx + d$ is divided by $3x - 2$, the quotient is $2x^2 + 2x + 3$ and remainder is 7 Find the values of a, b, c, and d and place them in their respective boxes starting on the left.

15. When a polynomial $P(x)$ is divided by $x - 4$, the remainder is 6. If a new polynomial $A(x) = 2P(x)$ is divided by $x - 4$, the remainder will be?

16. When $x^3 - 4x^2 + 3$ and $x^3 - 3x^2 - 8x + 19$ are each divided by $x - a$, the remainders are equal. What is the value of a?

3 The Factor Theorem

Opener: When $x^3 + kx^2 - 2x - 7$ is divided by $x + 1$, the remainder is 5. What is the remainder when it is divided by $x - 1$?

Factor Theorem

If $P(x)$ is a polynomial, then $P(x)$ has a factor $x - a$, if and only if, $P(a) = 0$. "If and only if" implies two things:

1. If $P(a) = 0$, then $x - a$ is a factor of $P(x)$.
2. If $x - a$ is a factor of $P(x)$, then $P(a) = 0$.

Example 3.1. Given $P(x) = 3x^3 + 4x^3 - 3x^2 - 3x - 10$, determine if $(x + 2)$ is a factor of $P(x)$.

Example 3.2. Given $P(x) = 3x^3 + 4x^3 - 3x^2 - 3x - 10$, determine if $(x - 1)$ is a factor of $P(x)$.

Factor Fully by Factor Theorem

1. Use the factor theorem to find one factor by testing numbers that are factors of the constant term.
2. Use the factor from Step 1 and synthetic or long division to find the remaining polynomial.
3. Continue factoring if possible.
4. Write your answer with all the factors together.

Example 3.3. What are the possible integral zeros of the polynomial $P(x) = x^4 - 2x^3 - 8x^2 + 3x - 4$?

Example 3.4. Fully factor the following.

a) $f(x) = x^3 - 9x^2 + 20x - 12$

b) $f(x) = 4x^3 + 12x^2 + 5x - 6$

Example 3.5. Solve the following equations by factoring.

a) $2x^3 - 3x^2 - 10x + 3 = 0$

b) $x^4 + 4x^3 - 7x^2 - 34x - 24 = 0$

Practice Exercises

1. What is the corresponding binomial factor of a polynomial, $P(x)$, given the value of the zero?

   a) $P(2) = 0$ b) $P(-4) = 0$ c) $P(b) = 0$ d) $P(-d) = 0$

2. Determine whether $x + 1$ is a factor of each polynomial.

   a) $x^3 + x^2 - x - 1$ b) $x^4 - 3x^3 - 4x^2 + x + 1$

   c) $2x^3 - x^2 - 3x - 1$ d) $4x^4 + 7x + 3$

3. What are the possible integral zeroes of each polynomial?

   a) $x^3 - 3x^2 + 4x - 16$ b) $x^3 + 2x^2 + 8x + 12$

4. Factor fully.

   a) $x^3 - x^2 - 4x + 4$ b) $x^3 - 2x^2 - 4x + 8$

   c) $x^4 + 2x^3 - x - 2$ d) $4x^3 - 8x^2 + x + 3$

e) $6x^3 + x^2 - 31x + 10$

f) $x^4 + x^3 - 13x^2 - 25x - 12$

5. What are all values of k for which $\frac{1}{2}$ is a zero of $P(x) = -4x^3 + 2x^2 - 2kx + k^3$

6. If $x = c$ is a root of the polynomial equation $P(x) = 0$, then what must be a factor of $P(x)$?

7. If $x - a$ is a factor of $2x^3 - ax^2 + (1 - a^2)x + 5$, what is the value of a?

8. Find the complete factored form of the following: $P(x) = x^5 + 3x^4 - 5x^3 - 15x^2 + 4x + 12$ if $P(-2) = P(-1) = P(1) = 0$.

9. Determine the values for a and b such that $x - 1$ is a factor of both $x^3 + x^2 + ax + b$ and $x^3 - x^2 - ax + b$.

10. Determine all real zeros of the function $P(x) = -2x^2(x^2+4)(x^2-4)$

    a) $0, \pm\sqrt{2}$

    b) $0, \pm 2$

    c) $0, \sqrt{2}, 2$

    d) $0, \pm\sqrt{2}, \pm 3$

11. If $P(x) = (x-3)q(x) + 3$, determine $P(3)$.

    a) $q(3)$

    b) $q(-3)$

    c) $-r$

    d) r

12. Given the synthetic division below, determine the product of $p \cdot q \cdot r$.

-3	2	3	p	r
		-6	q	-12
	2	-3	4	0

13. For $P(x) = x^2 - x + 1$, find a, if $a > 0$ and $P(a) = 3$.

14. If $-5x$ is a factor of the polynomial $P(x)$, then $P(0)$ is

    a) -5 b) $\frac{1}{5}$ c) 0 d) 5

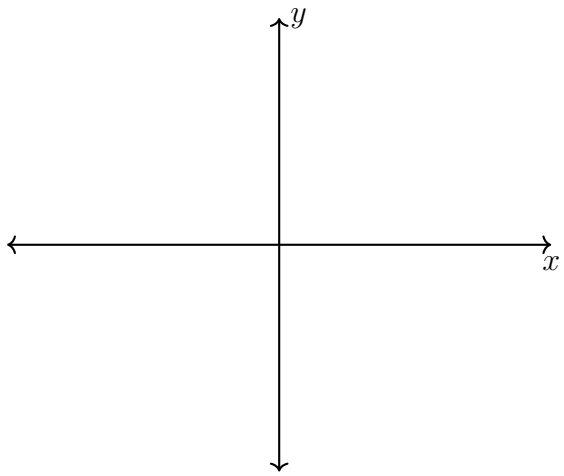
15. If $x - c$ is a factor of $6x^3 + 3cx^2 - c^2x - 27$, then the value of c correct to the nearest tenth is?

16. The y-intercept of the graph of the function defined by $P(x) = a(x-r_1)(x-r_2)(x-r_3)$ is

    a) $r_1 + r_2 + r_3$ b) $r_1 \cdot r_2 \cdot r_3$ c) $-a \cdot r_1 \cdot r_2 \cdot r_3$ d) $-a(r_1+r_2+r_3)$

4 Equations and Graphs of Polynomial Functions

Opener: Factor $y = x^3 - 3x^2 - 34x + 120$ and determine the x and y-intercepts. Sketch the graph.



Positive and Negative Intervals

A function is positive ($f(x) > 0$) where it lies _____ the x-axis.

A function is negative ($f(x) < 0$) where it lies _____ the x-axis.

Example 4.1. For the function above $y = x^3 - 3x^2 - 34x + 120$, identify the intervals where it is positive and negative.

Interval Notation

Example 4.2. Write the following in ***interval notation***.

a) $5 < x < 10$ b) $x \leq -1$ c) $-2 \leq x < 7$

Multiplicity

The **multiplicity** is the number of times a zero of a polynomial function occurs.

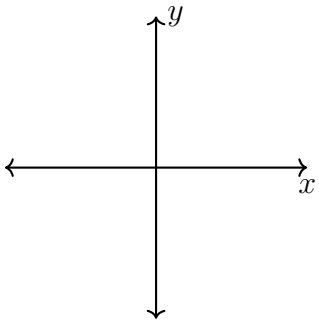
The polynomial $(x-2)(x-2)(x-2)(x+1)(x+1)$ has a multiplicity of ____, at $x=2$.

Because the factor $(x-2)$ occurs ____ times, it is odd/even.

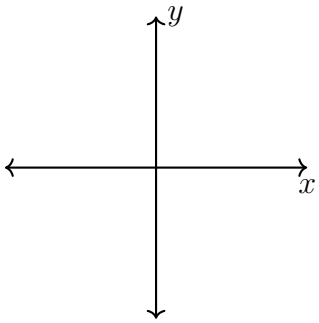
The zero at $x=-1$, has a multiplicity of ____ because the factor $(x+1)$ occurs ____ times, making it odd/even.

Example 4.3. Sketch the following and determine the rules that relate to multiplicity.

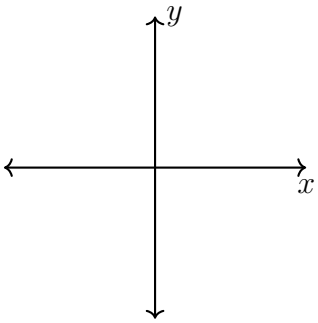
a) $y=(x-1)$



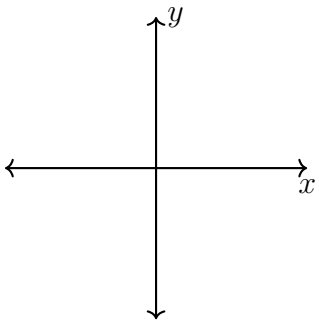
b) $y=(x-1)^2$



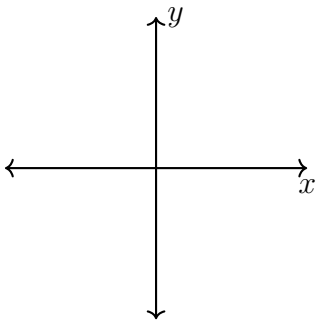
c) $y=(x-1)^3$



d) $y=(x-1)^4$



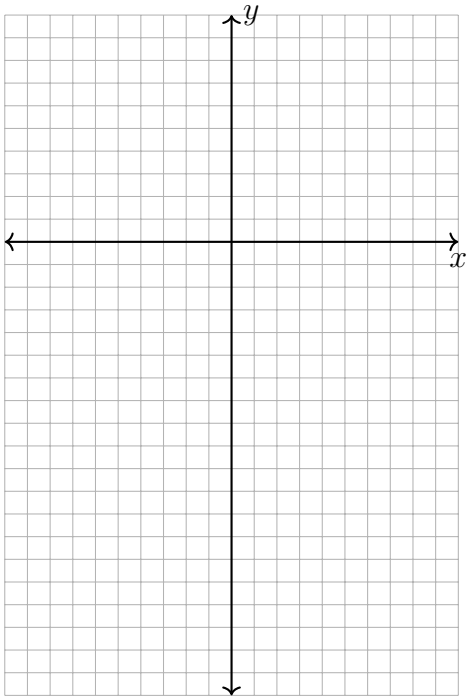
e) $y=(x-1)^5$



General rule(s) relating to multiplicity

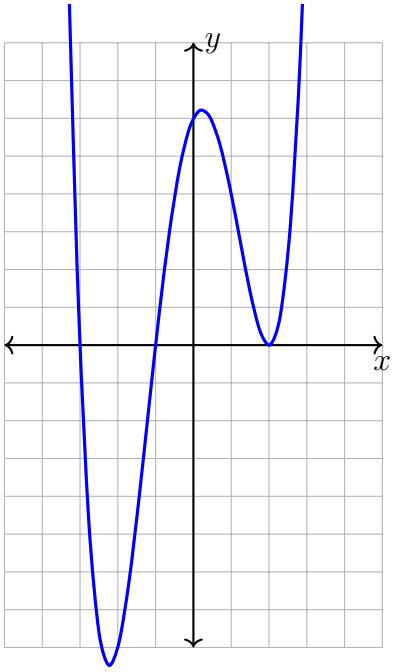
Example 4.4. Factor the polynomial function $y = x^3 + x^2 - 8x - 12$, complete the table and sketch the function without technology.

Degree	
Leading Coefficient	
End behaviour	
Zeros/x-intercepts/roots	
Multiplicity	
y-intercept	

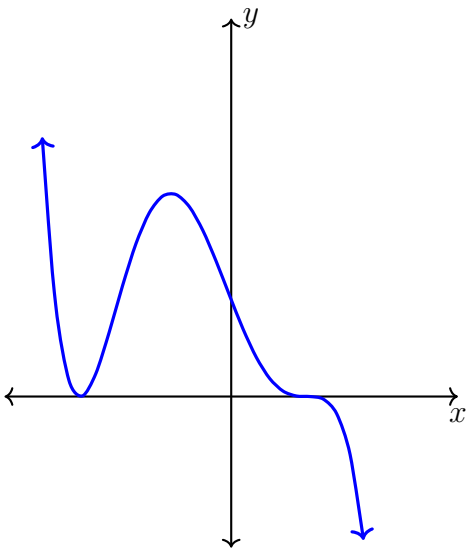


Example 4.5. Use the graph of the given functions to write the corresponding polynomial equations:

a)



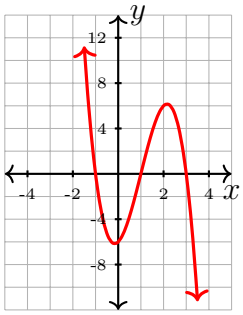
b)



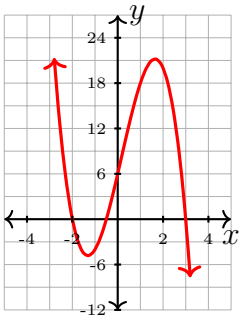
Practice Exercises

1. Use the graph of the given function to write the corresponding polynomial equation in factored form.

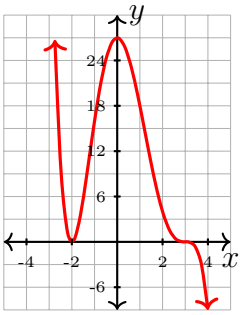
a)



b)

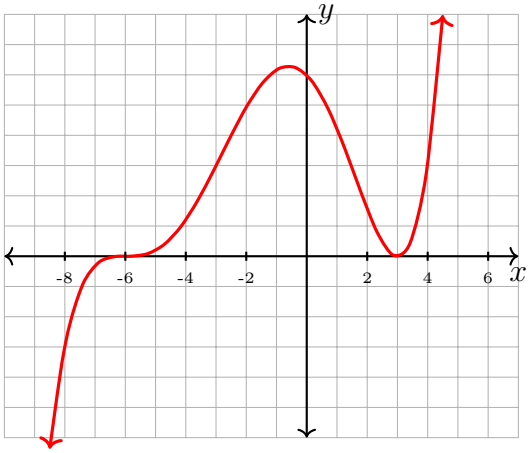


c)

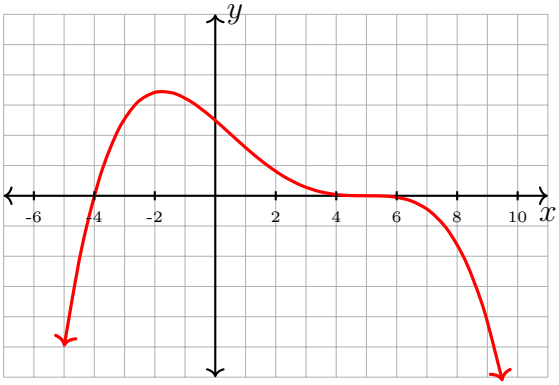


2. For each of the following graphs, state the factors and the multiplicity for each

a)



b)



3. Determine the equation with least degree for each polynomial function.

   a) A quartic function with zeros 2 (multiplicity of 3) and -5, and y-intercept 30.

   b) A quintic function with zeros -1 (multiplicity 2), 3 (multiplicity 1) and -2 (multiplicity 2), and constant term -12.

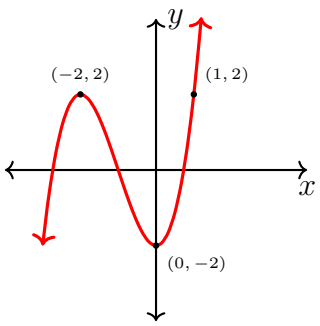
4. The graph of a cubic polynomial function $y = f(x)$ is shown. Determine the equation of $g(x) = f(x) - 2$

   a) $g(x) = (x+2)^2(x+1)$

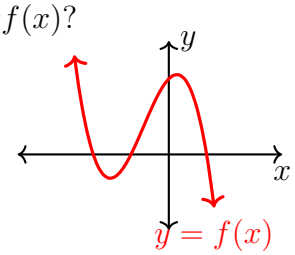
   b) $g(x) = (x+2)^2(x-1)$

   c) $g(x) = 2(x+2)^2(x+1)$

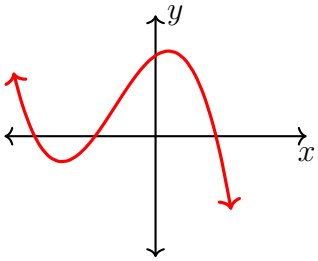
   d) $g(x) = 2(x+2)^2(x-1)$



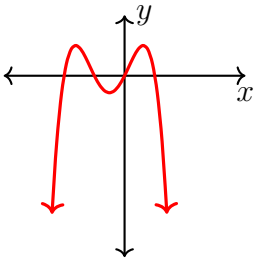
5. Given the graph of $y = f(x)$, which graph best represents $y = xf(x)$?



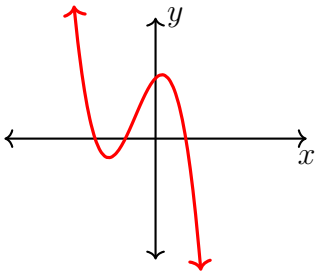
a)



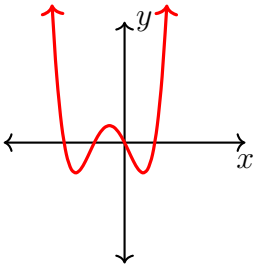
b)



c)



d)



6. The following table was used to graph a third degree polynomial

x	-2	-1	0	1	2	3
$P(x)$	-16	0	4	2	0	4

The third degree polynomial function that was graphed was

a) $P(x) = (x-2)(x+1)^2$

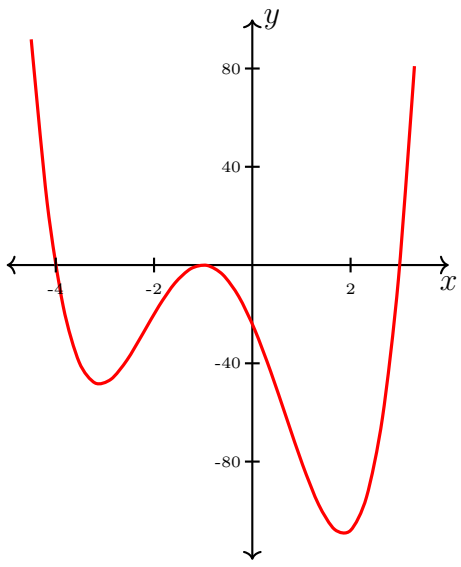
b) $P(x) = (x-2)^2(x+1)$

c) $P(x) = (x+2)^2(x-1)$

d) $P(x) = (x+2)(x-1)^2$

Use the following information to answer the next question

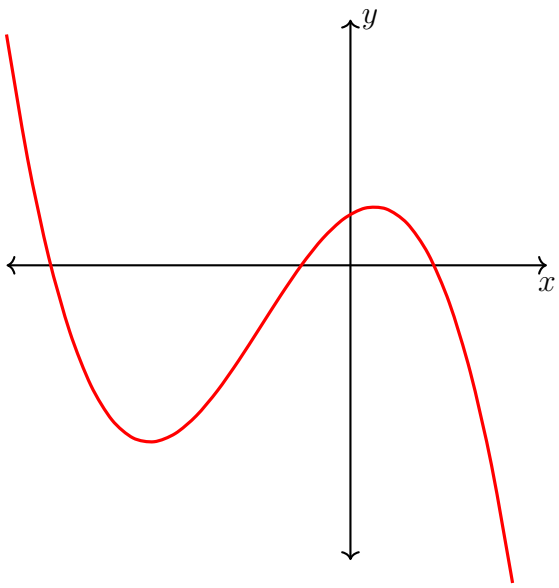
The partial graph of a fourth-degree polynomial function f, is shown below. The x-intercepts are $-4, -1$, and 3. The y-intercept is -24.



7. If $f(x) = a(x+b)^2(x+c)(x-d)$, where $a, b, c, d > 0$, find the values of a, b, c, d.

Use the following information to answer the next question

A student attempted to enter the equation $y = ax^3 - bx^2 + cx + d$ (where a, b, c, d are all positive integers) into their graphing calculator and arrived at the following graph.



8. The graph does **not** represent the graph of $P(x)$ because the student entered **one** term of the equation incorrectly. Which term was entered incorrectly?

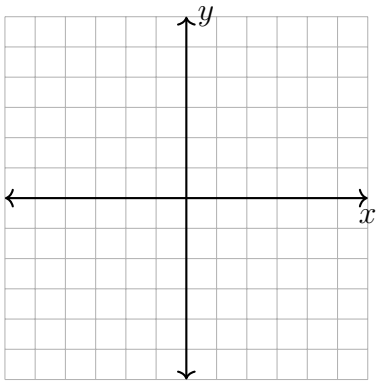
a) ax^3 b) $-bx^2$ c) cx d) d

Review Set A

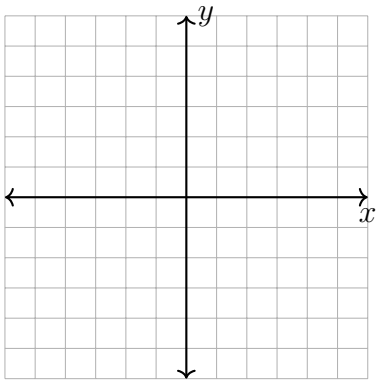
1. For each polynomial, find:

   i. zeros and their multiplicities
   ii. y-intercept
   iii. the end behaviour
   iv. sketch the graph without a graphing calculator

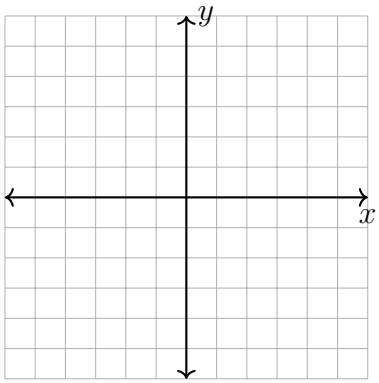
   a) $P(x) = (x-1)^2(x+2)^2$



   b) $P(x) = -(2x-1)(2x+1)$



   c) $P(x) = x^3 - 4x^2 - 25x + 100$

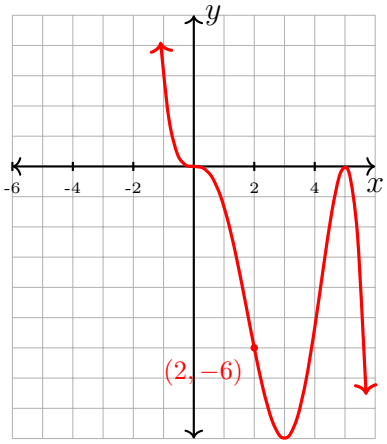


2. Determine the equation of the polynomial in factored form:

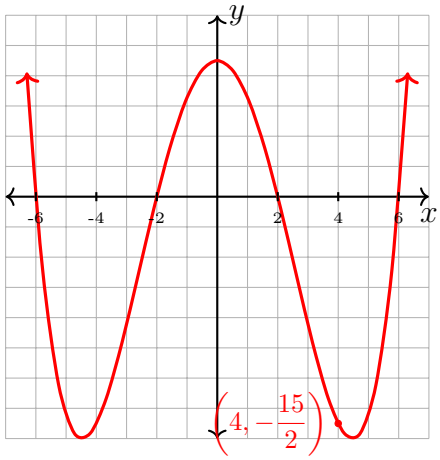
   a) If the zeros are $-1, -1, 2$ and the y-intercept is -4.

   b) A degree of 4 that has $-\frac{1}{2}$ as a root of multiplicity 3, and $2x^2 - x - 1$ as a factor.

   c)



   d)



3. Divide the following

   a) $x^3 + 9x^2 + 3$ by $x - 2$, using long division. Express your answer in $\frac{P(x)}{x-a} = Q(x) + \frac{R}{x-a}$ form.

   b) $12x^3 + 13x^2 - 23x + 7$ by $x + 1$, using synthetic division. Express your answer in $P(x) = DQ + R$ form.

4. When $P(x) = 2x^3 - 4x$ is divided by $x - 3$, what is the remainder?

5. When $x^3 + ax^2 + 3x + 2$ is divided by $x + 2$, the remainder is 32. What is the value of a?

6. Algebraically determine the roots of the polynomial $P(x) = 2x^3 + 7x^2 + 2x - 3$.

Review Set B

1. When the polynomial $P(x) = x^3 + 6x^2 - 3x - 3m$ is divided by $x + 3$, the remainder is 12. What is the value of m?

   A. 3 B. 8
   C. 10 D. 24

2. When $4x^3 - 2x^2 + 4x + m$ is divided by $2x - 3$, the quotient is $2x^2 + 2x + 5$ and the remainder is 22. What is the value of m?

   A. -15 B. 7
   C. 15 D. 37

3. Two factors of $2x^3 - mx + n$ are $(x + 2)$ and $(x - 1)$. What are the values of m and n respectively?

   A. 3 and 10 B. 4 and 8
   C. 5 and 6 D. 6 and 4

4. If a polynomial, $P(x)$, is divided by $x + a$, the remainder R is equal to

   A. $P(-a)$ B. $P(a)$
   C. a D. 0

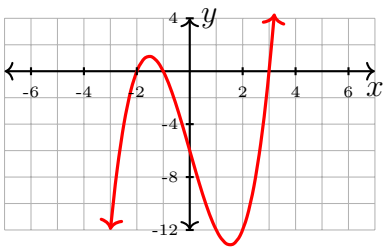
5. If $P(4) = 0$, then a complete factorization of $P(x) = 3x^3 - 17x^2 + 18x + 8$ is in the form $(x - 4)(ax + b)(x - c)$. Find the values of a, b, and c.

6. $P(-5) = P(-1) = 0$, then a second degree factor of the polynomial $P(x)$ is

   A. $x^2 - 6x - 5$ B. $x^2 - 6x + 5$
   C. $x^2 + 6x - 5$ D. $x^2 + 6x + 5$

7. When $P(x) = x^3 + kx^2 - x - 30$ is divided by $x + 2$, the remainder is -12. If $P(x)$ is divided by $x - 3$, find the remainder.

8. The graph of the polynomial function $y = f(x)$ is shown below.

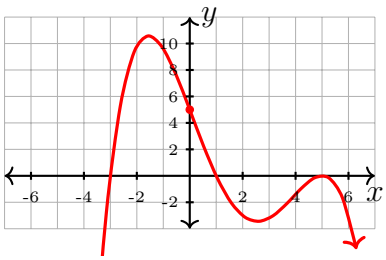


A possible equation for the graph below could be?

A. $x^3 + 6$ B. $x^3 - 7x - 6$

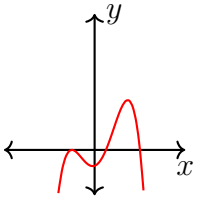
C. $x^3 + 7x - 6$ D. $x^3 - 7x^2 + 6x + 6$

9. The graph of the polynomial function $y = f(x)$ is shown below.



The polynomial above can be written in the form $y = -\frac{1}{a}(x + b)(x - c)(x - d)^e$. Find the value of a.

10. Karen makes the following statements about the graph of the polynomial function to the right.



i. The range of the function is $\{y \mid y \geq max\,value\}$
ii. The constant term of the corresponding function is negative
iii. There are 3 zeros
iv. Since the graph extends down into quadrant 3 and quadrant 4, the leading coefficient of the polynomial function is negative.

Which of the statements are correct?

A. i and ii B. ii and iii C. ii, iii, and iv D. all of them

11. Given the polynomial $y = 3x^3 + 11x^2 + 8x - 4$; choose the following statements that best describe the characteristics.

    1. Domain is $x \in \mathbb{R}$
    2. Range is $y \geq -4$
    3. The function has a multiplicity of 2 at the factor $(x + 2)$
    4. The y-intercept is at $y = -4$
    5. The point $(-1, 4)$ exists on the graph
    6. The remainder when the polynomial is divided by $(x + 5)$ is -144
    7. The polynomial can be factored as $y = (x + 2)^2(3x + 1)$
    8. The end behaviour of the graph approaches $-\infty$ when $x \to \infty$

    The number of true statements is:

12. If $P(x)$ is a polynomial function where $P\left(-\frac{2}{3}\right) = 0$ and $P(0) = 12$, then ___i___ is a factor of $P(x)$ and ___ii___ is a constant term in the equation $P(x)$.

    The statement is completed by the information in row

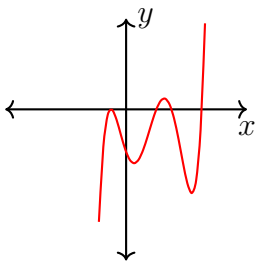
Row	i	ii
A.	$(3x - 2)$	12
B.	$(3x + 2)$	12
C.	$(3x - 2)$	-12
D.	$(3x + 2)$	-12

13. Given that $x = 3$ is a root of $2x^3 - 3x^2 - 10x + 3 = 0$, the exact value of the other roots can be written in the form $x = \dfrac{-a \pm \sqrt{b}}{c}$. The value of $a + b + c$ is

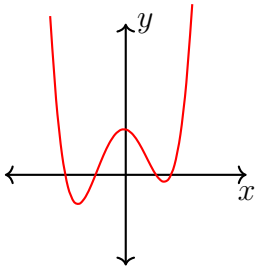
14. The graph of $P(x) = 3x^4 + 14x^3 + x^2 + kx + d$ has a root at $x = -2$ and a y-intercept of 8. Which of the following is also a root of $P(x)$?

    A. $x = -1$ B. $x = 2$ C. $x = 3$ D. $x = -4$

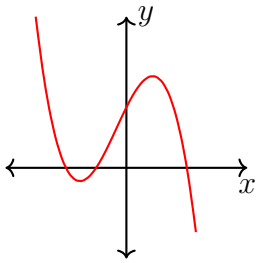
15. Use the graphs below to answer the statements



Graph 1



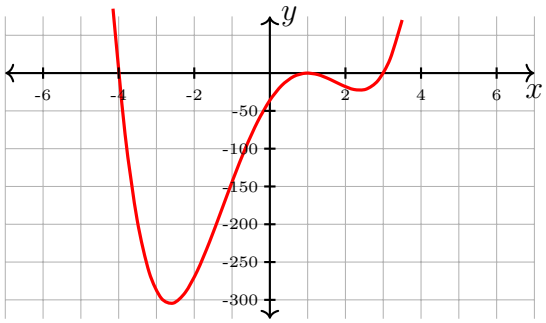
Graph 2



Graph 3

The graph that represents a polynomial function with a negative constant term is ___
The graph that represents a polynomial function with a negative leading coefficient is ___
The graph that represents a polynomial function with a zero of multiplicity 2 is ___
The graph that represents a polynomial function with a degree of 4 is ___

16. The partial graph of the polynomial $P(x)$ shown below has a y-intercept of $(0, -36)$ and can be expressed as a polynomial function of the form $P(x) = a(x+b)(x-c)(x-d)^2$



The value of a is _____

The value of d is _____

17. The following **True** statements are: _____, _____, and _____.

1. All even degree polynomial functions have at least one x-intercept.
2. All odd degree polynomial functions have at least one x-intercept.
3. The range of an even degree function is always $y \in \mathbb{R}$.
4. If the leading coefficient of a polynomial function is negative, as the x-values increase, the $y-$ values also increase.
5. The range is determined by the constant term of the polynomial.
6. The remainder when $P(x)$ is divided by $x - a$ is $P(a)$.
7. The polynomial function $P(x)$ has a factor of $x + a$ if $P(a) = 0$.
8. The constant term of the polynomial function is the y-intercept of the graph.

Answer Key

Section 1 - Exercises

1. a) **Graph D**
   b) **Graph B**
   c) **Graph A**
   d) **Graph C**

2.

Polynomial Function	Degree	Type	Leading Coefficient	Constant Term
$f(x) = 6x^3 - 5x^2 + 2x - 8$	**3**	**Cubic**	**6**	**-8**
$y = -2x^5 + 5x^3 + x^2 + 1$	**5**	Quintic	**-2**	**1**
$g(x) = x^3 - 7x^4$	**4**	**Quartic**	**-7**	**0**
$p(x) = 10x - 9$	**1**	**Linear**	**10**	**-9**
$y = -0.5x^2 + 4x + 3$	**2**	**Quadratic**	**-0.5**	**3**
$h(x) = 3x^4 - 8x^3 + x^2 + 2$	**4**	**Quartic**	**3**	**2**
$y = -5$	**0**	Constant	**-5**	**-5**

3. a) **End Behaviour:** QII to QIV
   x**-intercepts:** 1 to 5
   y**-intercept:** 0
   b) **End Behaviour:** QII to QI
   x**-intercepts:** 0 to 4
   y**-intercept:** -1

4. a) Graph starts in QIII (bottom left) and ends in QI (top right). Crosses or touches the x-axis exactly 2 times.
   b) Graph starts in QIII (bottom left) and ends in QIV (bottom right). Crosses the x-axis exactly 4 times.

5. **b, d, f, g**

6. a) **15, 000**
   b) **18, 000**
   c) In approximately **18 years** (At $t = 18$, the population is $23,856$)

7. a) **End Behaviour:** QIII to QI
   x**-intercepts:** Up to 5
   y**-intercept:** 0
   b) **End Behaviour:** QIII to QIV
   x**-intercepts:** Up to 4
   y**-intercept:** 0

8. **c** ($a < 0, d \neq 0$)
9. **c** ($a > 0, e < 0$)
10. **b** (1)
11. **c** (ii, iii, iv)
12. **d** ($f(x) = -4$)
13. **c** (3)

Section 2 - Exercises

1. a) Quotient: 23, Remainder: 17 b) Quotient: 29, Remainder: 6

2. $x^3 + 3x^2 - 2x + 5 = (x + 1)(x^2 + 2x - 4) + 9$

3. a) $2x^3 + x^2 - 27x - 36 = (x + 3)(2x^2 - 5x - 12) + 0$
   b) $3x^4 + 2x^3 - 6x + 1 = x(3x^3 + 2x^2 - 6) + 1$
   c) $3x^4 - x^3 - 5 = (x - 3)(3x^3 + 8x^2 + 24x + 72) + 211$
   d) $-4x^4 + 11x - 7 = (x - 3)(-4x^3 - 12x^2 - 36x - 97) - 298$

4. a) -36 b) 8

5. a) -9 b) -2
   c) -4

6. $m = 3$

7. **a** $(P(-4))$

8. **c** $(P(a) = b)$

9. **b** $((-3, 4))$

10. **a** (-4)

11. **a** $(a = -7, b = -6)$

12. **a** $(a = -22, b = -4)$

13. **d** $(-3, -7)$

14. $6, 2, 5, 1$

15. 12

16. $a = 4$

Section 3 - Exercises

1. a) $x - 2$ b) $x + 4$ c) $x - b$ d) $x + d$

2. a) Yes $(P(-1) = 0)$ b) Yes $(P(-1) = 0)$
   c) No $(P(-1) = -1)$ d) Yes $(P(-1) = 0)$

3. a) $\pm 1, \pm 2, \pm 4, \pm 8, \pm 16$ b) $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12$

4. a) $(x-2)(x+2)(x-1)$ b) $(x-2)^2(x+2)$
   c) $(x-1)(x+2)(x^2+x+1)$ d) $(x-1)(2x-3)(2x+1)$
   e) $(x-2)(3x-1)(2x+5)$ f) $(x+1)^2(x-4)(x+3)$
5. $k=-1,0,1$
6. $x-c$
7. $a=-5$
8. $(x-1)(x+1)(x-2)(x+2)(x+3)$
9. $a=-1, b=-1$
10. **b** $(0,\pm 2)$
11. **d** (r)
12. -540 *(since $p=-5, q=9, r=12$)*
13. $a=2$
14. **c** (0)
15. 1.5 *(since $8c^3=27$)*
16. **c** $(-a\cdot r_1\cdot r_2\cdot r_3)$

Section 4 - Exercises

1. a) $y=-2(x+1)(x-1)(x-3)$ b) $y=-(x+2)(2x+1)(x-3)$ c) $y=-\frac{1}{4}(x+2)^2(x-3)^3$
2. a) $(x+6)$ mult 3, $(x-3)$ mult 2 b) $(x+4)$ mult 1, $(x-5)$ mult 3
3. a) $P(x)=-\frac{3}{4}(x-2)^3(x+5)$ b) $P(x)=(x+2)^2(x+1)^2(x-3)$
4. **b** $(g(x)=(x+2)^2(x-1))$
5. **b** *(Multiplying by x turns the cubic into a quartic. Since the original cubic ends in QIV, the leading coefficient is negative, meaning the new quartic will open downward. The x factor adds a new x-intercept at 0, making the correct graph cross at -2, -1, 0, and 1.)*
6. **b** $(P(x)=(x-2)^2(x+1))$
7. $a=2, b=1, c=4, d=3$ *(The graph is tangent at -1, so the squared factor is $(x+1)^2 \implies b=1$. It crosses at -4 and $3 \implies c=4, d=3$. The y-intercept is $-24 \implies a(1)^2(4)(-3)=-24 \implies -12a=-24 \implies a=2$.)*
8. **a** (ax^3) *(If a is a positive integer, the leading coefficient is positive, meaning the graph should end in Quadrant I (top right). The graphed function ends in Quadrant IV, meaning the student likely entered a negative value for a.)*

Review Set A

1. a)
   i. **Zeros:** $x = 1$ (multiplicity 2), $x = -2$ (multiplicity 2)
   ii. **y-intercept:** 4
   iii. **End behaviour:** QII to QI (As $x \to \pm\infty, y \to \infty$)
   iv. **Sketch:** A "W" shaped graph touching (bouncing off) the x-axis at -2 and 1, crossing the y-axis at 4.

   b)
   i. **Zeros:** $x = \frac{1}{2}$ (multiplicity 1), $x = -\frac{1}{2}$ (multiplicity 1)
   ii. **y-intercept:** 1
   iii. **End behaviour:** QIII to QIV (As $x \to \pm\infty, y \to -\infty$)
   iv. **Sketch:** A downward-opening parabola crossing the x-axis at -0.5 and 0.5, and the y-axis at 1.

   c)
   i. **Zeros:** $x = 5$ (multiplicity 1), $x = -5$ (multiplicity 1), $x = 4$ (multiplicity 1) *(from factored form: $(x-5)(x+5)(x-4)$)*
   ii. **y-intercept:** 100
   iii. **End behaviour:** QIII to QI (As $x \to -\infty, y \to -\infty$ and $x \to \infty, y \to \infty$)
   iv. **Sketch:** A cubic curve starting bottom left, crossing the x-axis at -5, rising steeply to cross the y-axis at 100, turning back down to cross the x-axis at 4, and turning up to cross at 5.

2. a) $P(x) = 2(x+1)^2(x-2)$
   b) $P(x) = a(2x+1)^3(x-1)$ or $P(x) = a(x+\frac{1}{2})^3(x-1)$ *(any non-zero a is correct)*
   c) $P(x) = -\frac{1}{12}x^3(x-5)^2$
   d) $P(x) = \frac{1}{32}(x+6)(x+2)(x-2)(x-6)$

3. a) $\dfrac{x^3+9x^2+3}{x-2} = x^2+11x+22+\dfrac{47}{x-2}$
   b) $12x^3+13x^2-23x+7 = (x+1)(12x^2+x-24)+31$

4. 42 *(using Remainder Theorem: $P(3) = 2(3)^3 - 4(3) = 54 - 12$)*

5. $a = 11$ *(using Remainder Theorem: $P(-2) = 32 \implies -8+4a-6+2 = 32 \implies 4a - 12 = 32$)*

6. $x = -3, x = -1, x = \frac{1}{2}$ *(Testing factors of -3 gives $P(-1) = 0$. Dividing by $(x+1)$ yields $2x^2+5x-3$, which factors to $(2x-1)(x+3)$)*

Review Set B

1. **B** (8)
2. **B** (7)
3. **D** (6 and 4)
4. **A** ($P(-a)$)
5. 3, 1, 2 *(from $a = 3, b = 1, c = 2$)*
6. **D** ($x^2 + 6x + 5$)
7. 48
8. **B** ($x^3 - 7x - 6$)
9. 15 *(so $a = 30$)*
10. **C** (ii, iii, and iv)
11. 4 *(Statements 1, 3, 4, and 6 are true)*
12. **B** (Row B)
13. 24 *($a = 3, b = 17, c = 4 \implies 3 + 17 + 4 = 24$)*
14. **D** ($x = -4$)
15. Graph **1**, Graph **3**, Graph **1**, Graph **2** (in that order)
16. The value of a is **3**
    The value of d is **1**
17. **2, 6, 8**